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Associations of Atrial Fibrillation after Non-Cardiac Surgery with Stroke, Subsequent Arrhythmia, and Mortality: Cohort Study

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Abstract

Background: Postoperative atrial fibrillation (AF) after non-cardiac surgery confers increased risks of ischemic stroke and transient ischemic attack (TIA). How outcomes for postoperative AF after non-cardiac surgery compare to AF occurring outside of the operative setting is unknown.

Objective: To compare the risks of ischemic stroke or TIA and other outcomes in patients with postoperative AF compared to those with incident AF not associated with surgery.

Design: Cohort study.

Setting: Olmsted County, MN.

Participants: Patients with incident AF between 2000 and 2013.

Measurements: Patients were categorized as AF occurring within 30 days of a non-cardiac surgery (postoperative AF) and those with AF unrelated to surgery (non-operative AF).

Results: Of 4,231 patients with incident AF, 550 (13%) had postoperative AF as their first-ever documented AF presentation. Over a mean follow-up of 6.3 years, 486 patients experienced an ischemic stroke or TIA, 2,462 had subsequent AF, and 2,565 deaths occurred. The risk of stroke or TIA was similar between those with postoperative AF and non-operative AF (absolute risk

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Author Contributions

Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved: AMC.

Substantial contributions to the conception or design of the work: KCS, BJG, AMC.

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difference (ARD) at 5 years, 0.1% [−2.9%–3.1%]; hazard ratio (HR), 1.01 [0.77–1.32]). A lower risk of subsequent AF was observed for patients with postoperative AF (ARD at 5 years, −13.4% [−17.8%- −9.0%]; HR, 0.68 [0.60–0.77]. Finally, no difference was observed for cardiovascular death or all-cause death between patients with postoperative AF and non-operative AF.

Limitations: The population consisted predominantly of whites; caution should be used extrapolating the results to more racially diverse populations.

Conclusions: Postoperative AF after non-cardiac surgery is associated with similar risk of thromboembolism compared to non-operative AF. Our findings have potentially important implications for the early post-surgical and subsequent management of postoperative AF.

Introduction

New-onset atrial fibrillation (AF) may occur sporadically or may be triggered by other factors, such as the perioperative state. AF occurring after non-cardiac surgery may be triggered by perioperative stress and systemic inflammation in patients with predisposing comorbidities (1). We recently demonstrated in a community-based cohort of patients undergoing non-cardiac surgery that in those who developed first documented AF within 30 days of the surgery, AF often recurred during subsequent follow-up and carried increased risks of thromboembolism and death compared to patients who had surgery but did not develop AF (2). However, it is less clear how postoperative AF after non-cardiac surgery compares to AF occurring outside of the operative setting with respect to subsequent risk of non-fatal and fatal outcomes. In this community-based cohort study, we sought to compare the risks of ischemic stroke or transient ischemic attack (TIA), subsequent AF, and mortality in patients with postoperative AF versus those with incident AF not associated with surgery.

Methods

Patient Population

This study utilized the Rochester Epidemiology Project (REP) medical records-linkage system (3–6). The REP captures nearly all medical record data from persons residing in Olmsted County, MN who provided authorization to use their medical records for research (obtained from 98% of the population). The Mayo Clinic and Olmsted Medical Center Institutional Review Boards approved this study.

Details on the study population and cohort creation have been previously published (2, 7). Briefly, the study population consisted of adult residents (≥18 years) of Olmsted County, MN with incident AF diagnosed between 1/1/2000 and 12/31/2013. Patients were identified using inpatient and outpatient International Classification of Diseases, Ninth Revision (ICD-9) diagnostic codes 427.31 and 427.32 from all health care providers in Olmsted County, and Mayo Clinic electrocardiograms (ECGs) indicating AF or atrial flutter. All records of patients with possible AF diagnoses were manually reviewed by trained nurse abstractors to validate the diagnosis. Qualifying diagnoses included a first-ever (incident) AF or atrial flutter event on one or more of (1) ECG or rhythm strip, (2) Holter monitor, event monitor, or telemetry, (3) monitor during an emergency department visit or hospitalization, (4) ECG during an echocardiogram, (5) pacemaker interrogation, or (6) physician diagnosis

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(2, 7). Patients with AF occurring within 30 days of heart or large thoracic vessel surgery were excluded, while major thoracic/pulmonary surgeries (non-bronchoscopic) were included.

Postoperative AF was defined as AF occurring intra-operatively or within 30 days after a non-cardiac surgical procedure. The surgical procedures were identified using ICD-9 procedure codes and Current Procedural Terminology (CPT) codes as defined by the Clinical Classifications Software (CCS) developed by the Agency for Healthcare Research and Quality (8). We included any inpatient or outpatient non-cardiac surgery (open or laparoscopic) performed for diagnostic or therapeutic purposes under general anesthesia. We excluded minor procedures such as endoscopic ear, nose, and throat (ENT), gastrointestinal, urologic, or bronchoscopic procedures; percutaneous vascular or non-vascular procedures; as well as minor skin, dental, ENT, and ophthalmologic procedures with only minimal sedation or local anesthesia.

The comparator group consisted of patients whose first-ever incident AF event did not occur within 30 days of a surgical procedure (non-operative AF group). A flow diagram of the cohort is shown in Appendix Figure 1.

Data Collection and Outcomes Ascertainment

As previously defined (2, 7), we documented comorbidities diagnosed on or prior to the date of incident AF using ICD-9 diagnostic codes. We calculated the $CHADS_2$, CHA_2DS_2-VASc, and Charlson comorbidity index scores (9–11). In addition, outpatient anticoagulant (warfarin, heparin, and direct oral anticoagulants) and antiarrhythmic prescription data were obtained from 2004 forward from Mayo Clinic and Olmsted Medical Center. Only outpatient prescriptions were obtained because medications administered during hospitalizations were not available in the REP. Finally, we documented AF-specific procedures after the diagnosis of AF for the entire study period, including direct current (DC) cardioversions (CPT codes 92960, 92961; ICD-9 codes 99.61, 99.62; ICD-10 code 5A2204Z), atrioventricular (AV) node catheter ablations (CPT code 93650), and AF catheter ablations (CPT codes 93651, 93653, 93655–93657).

The primary outcome was the composite of ischemic stroke or TIA. Secondary outcomes included subsequent AF, all-cause death and cardiovascular-related death. Ischemic strokes and TIAs were identified using diagnostic codes (ICD-9 codes 433–436, ICD-10 codes I63 and G45), and were manually validated using the following criteria (12). For ischemic strokes, we required both: 1) acute onset (minutes to hours) of a focal neurologic deficit persisting >24 hours caused by altered circulation to a limited region of the cerebral hemispheres, brainstem, or cerebellum, and 2) no evidence of intracerebral hemorrhage on computed tomography (CT), magnetic resonance imaging (MRI), or autopsy (if available). For TIAs, we required either: 1) focal neurological symptoms with abrupt onset and rapid resolution, lasting <24 hours, and caused by focal cerebral ischemia, or 2) transient monocular visual disturbance with abrupt onset and rapid resolution, lasting <24 hours, and caused by retinal ischemia. Subsequent AF was identified using CPT codes and the Mayo Clinic ECG database. Manual review of the records was completed to validate subsequent AF, which was defined as ECG-confirmed AF documented >30 days after the index date.

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Deaths were identified using death certificates from the State of Minnesota, autopsy reports, and obituary notices. The underlying cause of death in the death certificate was used to define cardiovascular-related deaths using the American Heart Association Classification (ICD-9 codes 390–459, ICD-10 codes I00-I99) (13).

Statistical Analysis

Analyses were performed using SAS software, version 9.4 (SAS Institute Inc. Cary, NC) and R software, version 4.0.3 (R Foundation for Statistical Computing, Vienna, Austria). Baseline characteristics of the patients with postoperative AF and non-operative AF were reported as frequencies and percentages for categorical variables and means±SD for continuous variables. Patient characteristics were compared between groups using the chi-square test for categorical variables and the two-sample t-test for continuous variables. For each outcome, follow-up began at the date of incident AF for both groups and continued until they experienced the outcome event, last follow-up, or December 31, 2018, whichever occurred first. Cumulative incidence plots (1-Kaplan Meier) were constructed to visualize the occurrence of each outcome for patients with postoperative AF compared to those with non-operative AF. Similarly, cumulative incidence plots (1-Kaplan-Meier) were constructed to assess the time to first anticoagulation prescription and time to first antiarrhythmic drug prescription after AF diagnosis for each group of patients. Because baseline characteristics differed between patients with postoperative AF and those with non-operative AF, a propensity score was constructed to balance the characteristics between the two groups among those with complete data. The probability of postoperative AF was modeled with logistic regression and included the following predictors: age, age^2, male, age×male, white race, the natural logarithm of body mass index, and comorbid conditions. Inverse probability weights were calculated using the propensity score (14); stabilized weights were used to reduce extreme weights and preserve the original sample size. In addition, stabilized weights below the 1st percentile and above the 99th percentile were truncated to the 1st and 99th percentile, respectively (14). Standardized differences, defined as the difference in the means (for continuous variables) or proportions (for categorical variables) divided by the pooled standard deviation, were used to assess balance in baseline characteristics after weighting using the stabilized weights (15). A threshold of 0.1 was used for the absolute value of the standardized difference to support the assumption of balance (16). Weighted Cox proportional hazards regression models were used to examine the hazard ratios (HR), absolute risks and absolute risk differences (ARD) of each outcome for patients with postoperative AF compared to non-operative AF. Unadjusted models and models adjusting for age, sex, and the Charlson comorbidity index were utilized. In addition, weighted cumulative incidence plots, adjusted for age, sex and Charlson comorbidity index, were constructed. Furthermore, a model was constructed additionally adjusting for anticoagulant and antiarrhythmic drug prescriptions. In this model, the prescriptions were modeled as time-dependent variables, and the analysis was restricted to the subset of patients with index dates from 2004 forward due to the availability of the prescription data. Finally, the proportional hazards assumption was tested using the scaled Schoenfeld residuals and found to be valid.

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Results

Of the 4,231 patients with incident AF, 550 patients (13.0%) had postoperative AF as their first-ever documented AF presentation (Appendix Figure 1). Most postoperative AF events occurred within the first week after the surgery (82%; median time from surgery to AF diagnosis 2 (IQR 1–5) days). The types of surgical procedures were orthopedic [n=144 (26.2%)], gastrointestinal [n=128 (23.3%)], respiratory [n=113 (20.5%)], urogenital [n=38 (6.9%)], neurosurgical [n=35 (6.4%)], and other [n=92 (16.7%)].

The mean age (73.3 vs. 72.3 years) and proportion male (52.0% and 52.5%) were similar in the postoperative and the non-operative AF groups, respectively. Most comorbidities were more prevalent in patients with postoperative AF (Table 1). Importantly, the mean CHA_2DS_2-VASc score was higher in patients with postoperative AF compared to those with non-operative AF (3.6±1.8 vs. 3.3±2.0). After inverse probability weighting, all patient characteristics were balanced between the postoperative and the non-operative AF groups (all standardized differences <0.10; Table 1).

Among the 437 patients with postoperative AF starting in 2004 when outpatient prescription data became routinely available, 216 (49.4%) were prescribed an anticoagulant as compared to 1,699 (61.7%) in the non-operative AF group. A total of 379 of the 437 postoperative AF patients had a CHA_2SDS_2-VASc score of 2 or more, and of those, 191 (50.4%) were prescribed an anticoagulant; thus, half of the anticoagulant-eligible patients with postoperative AF never received an anticoagulant prescription. The median (IQR) time from AF diagnosis to first anticoagulant prescription was 37 (5–683) and 13 (2–283) days in the postoperative and non-operative AF groups, respectively. The cumulative incidence of anticoagulant prescriptions at 30 days and 1 year after the index AF diagnosis were 25.1% (95% CI, 20.7%–29.2%) and 39.1% (95% CI, 33.9%–43.8%), respectively, for the postoperative AF group, as opposed to 37.1% (95% CI, 35.3%–39.0%) and 50.9% (95% CI, 48.8%–52.8%), respectively, for the non-operative AF group ($p<0.01$; Figure 1). In the full cohort (550 patients with postoperative AF and 3,681 with non-operative AF), the cumulative incidence of electrical cardioversion procedures at 1 year after index was 9.6% (95% CI, 7.3%–12.2%) for the postoperative AF patients and 20.6% (95% CI, 19.2%–21.9%) for the non-operative AF patients. The cumulative incidence of catheter ablation procedures at 1 year was 0.7% (95% CI, 0.2%–1.6%) and 2.9% (95% CI, 2.3%–3.4%) for the postoperative and non-operative AF patients, respectively.

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Over a mean follow-up of 6.3±4.9 (range 0–19.0) years, 486 patients experienced an ischemic stroke or TIA, 2,462 patients had subsequent documented AF, and 2,565 deaths occurred, of which 1,017 (39.6%) were cardiovascular-related. The rate of subsequent AF was lower for patients with postoperative AF, but the rates of ischemic stroke or TIA were similar in the two groups (Table 2, Figure 2). Furthermore, although the rate of all-cause death was higher for patients with postoperative AF, the rates of cardiovascular-related death were similar in the two groups.

Using weighted analyses and after adjustment for age, sex, and Charlson comorbidity index, no difference in the risk of stroke/TIA was observed (ARD at 5 years, 0.1% [95% CI, −2.9%–3.1%]; HR, 1.01 [95% CI, 0.77–1.32]) for patients with postoperative AF vs. non-operative AF). However, a lower risk of subsequent AF was observed for patients with postoperative AF as compared to patients with non-operative AF (ARD at 5 years, −13.4% [95% CI, −17.8%- −9.0%]; HR, 0.68 [95% CI, 0.60–0.77]; Table 2, Figure 3, Appendix Figure 2). Finally, no differences in cardiovascular-related death or all-cause death were observed between patients with postoperative AF and non-operative AF in the adjusted analyses.

Similar patterns of association were observed after adjustment for age, sex, Charlson comorbidity index, and anticoagulant and antiarrhythmic drug prescriptions. Patients with postoperative AF had a similar risk of stroke/TIA (ARD at 5 years, 1.1% [95% CI, −2.2%–4.4%]; HR, 1.11 [95% CI, 0.81–1.51]), and a lower risk of subsequent AF (ARD at 5 years, −8.8% [95% CI, −13.7%- −3.9%]; HR, 0.77 [95% CI, 0.67–0.89]) compared to patients with non-operative AF.

During the period when prescription data were available (starting in 2004), 46 ischemic stroke/TIA events occurred among the postoperative AF group. Only 9 (19.6%) of these patients were prescribed an anticoagulant at the time of the stroke/TIA. In comparison, 114 of the 297 (38.4%) non-operative AF patients with an ischemic stroke/TIA during this time period were prescribed an anticoagulant at the time of the stroke/TIA.

Discussion

Our population-based study brings important data on associations of postoperative AF after non-cardiac surgery with outcomes. Our main findings are the following: 1) postoperative AF is common and comprises 13% of all new diagnoses of AF in the community; 2) the cumulative incidence of subsequent documented AF is approximately 18% at 1 year after the index periprocedural AF episode; 3) compared to AF unrelated to a surgical procedure, postoperative AF was associated with similar risks of stroke/TIA and death; 6) oral anticoagulation was underutilized in patients with postoperative AF.

Our study offers novel insights regarding the natural history of AF after non-cardiac surgery. We found that approximately 18% of patients with postoperative AF experienced subsequent documented episodes of AF within 1 year after the index postoperative AF episode. This estimate is similar to the reported AF recurrence of 16.8% during 2 years of follow-up in a study of patients undergoing noncardiac surgeries at Asan Medical Center (Seoul, Korea)

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(17). However, the actual recurrence rate of AF could be higher considering the frequency of subclinical AF. While subsequent documentations of AF were less common for patients with postoperative compared to non-operative AF, this finding may have important rhythm monitoring and treatment implications. In many patients, AF after non-cardiac surgery may not be a transient arrhythmia that is simply dependent on resolution of the perioperative state. Rather, the perioperative state may be considered a “stress test” for the manifestation of AF. A subset of the new diagnoses of postoperative AF may also represent previously undiagnosed primary AF that is detected due to more intensive rhythm monitoring in the perioperative period.

Recent evidence suggests an increased risk of thromboembolism and other adverse cardiovascular endpoints in patients experiencing AF after non-cardiac surgery. We recently reported the association of postoperative AF with stroke and TIA compared to patients without AF after similar types of surgeries (2). There was also a significant association with all-cause mortality and subsequent documented AF. Specifically, postoperative AF was associated with a 2.7-fold increased risk of stroke/TIA, 7.9-fold increased risk of subsequent AF, and 1.7-fold increased risk of death compared to patients without AF. These results suggest that patients with postoperative AF may require ongoing surveillance for the arrhythmia and its complications, though the optimal approach to follow-up and treatment is unknown. Similar conclusions had been previously reached from an administrative claims analysis from California hospitals (18), from a secondary analysis of the Perioperative Ischemic Evaluation Study (POISE) trials (19), and a recent meta-analysis of 18 studies reporting stroke and 17 studies reporting all-cause mortality related to non-cardiac postoperative AF (20).

The results of the current analysis further support the prognostic importance of postoperative AF through a direct comparison to patients experiencing their first-ever AF episode outside of the operative setting. In analyses adjusted for anticoagulant and antiarrhythmic drug use, we found no differences in the risk of stroke or death between the two types of AF despite fewer subsequent documentations of AF in the postoperative AF group. This result is consistent with a Danish nationwide study of administrative claims data where the outcomes of postoperative AF after non-cardiac surgery were compared to the outcomes of nonprocedural AF.(21) The strength of the current study is the manual validation of all AF diagnoses and outcome events rather than reliance on diagnostic codes alone. Collectively these findings further highlight that postoperative AF after non-cardiac surgery should not be considered a transient, benign phenomenon. Rather, it is an arrhythmia that appears to portend similar thromboembolic risks as AF diagnosed outside of the operative setting.

It is noteworthy that less than half of patients with postoperative AF were prescribed an anticoagulant during follow-up with a median time to anticoagulant prescription of more than 1 month from the index AF diagnosis. Among the patients with postoperative AF who experienced an incident stroke or TIA in this cohort, only 19.6% were receiving an anticoagulant at the time of the stroke or TIA event, compared with 38.4% of the non-operative AF patients who experienced a stroke or TIA. The underutilization of anticoagulation may reflect the perception that postoperative AF is an isolated, provoked arrhythmia after non-cardiac surgery that carries less severe implications than other forms

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of AF. Our data demonstrate that this perception may be erroneous and underscores a therapeutic gap with direct clinical relevance.

As well, it is possible that providers remain concerned of the risk of bleeding in the postoperative period when prescribing anticoagulants, which may also account for the low prevalence of anticoagulation prescriptions in patients with postoperative AF. The observational nature of our study does not permit any conclusions regarding the benefit of anticoagulation in the postoperative AF population. However, given the important thromboembolic risk implications of postoperative AF, our data support the consideration of anticoagulation in these patients in the absence of excessive bleeding risk. The complex decision-making regarding patient selection, timing and duration of anticoagulant therapy warrants further study in randomized trials.

Limitations

Certain limitations of our study should be acknowledged. First, the Olmsted County population consists predominantly of whites and extrapolating to more racially diverse populations should be done with caution. Second, we did not have information to categorize AF as paroxysmal or persistent AF. Third, although we performed weighted analyses to account for differences in baseline characteristics between the postoperative AF and non-operative AF groups and adjusted for demographics and comorbidities, it is possible that residual confounding in the associations exists due to unmeasured confounders. Fourth, our cohort was restricted to patients diagnosed with AF through 2013, and thus may not reflect more contemporary practice patterns, including prescribing of anticoagulation and utilization of rhythm control procedures. Finally, differences in follow-up between the postoperative AF and non-operative AF groups could have affected the results. The rates of ECG use (2.09 vs. 1.80 per person-year) and the rates of outpatient encounters (9.46 vs. 8.76 per person-year) during follow-up were higher for the postoperative AF group compared to the non-operative AF group. Thus, the observed lower rate of subsequent AF for the postoperative AF group is unlikely to be an artifact of the difference in follow-up between groups. Any potential bias in our results due to differences in follow-up would have most likely resulted in more conservative estimates, with the true association being more extreme than reported herein.

Conclusions

This community cohort study indicates that among patients undergoing non-cardiac surgery, AF occurring within 30 days of the surgery may be commonly a recurrent rather than transient arrhythmia. After adjustment for potential confounders, postoperative AF was associated with similar risks of stroke and TIA, and cardiovascular-related and all-cause death compared to other AF documented outside of the operative setting. Furthermore, oral anticoagulants were underutilized in patients with periprocedural AF. Our findings have potentially important implications that are pertinent to clinicians involved in the early post-surgical and follow-up care of patients in primary and subspecialty care settings. Randomized trials are required to establish the value of various management approaches

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for AF after non-cardiac surgery, such as rhythm monitoring, anticoagulation, and rhythm control.

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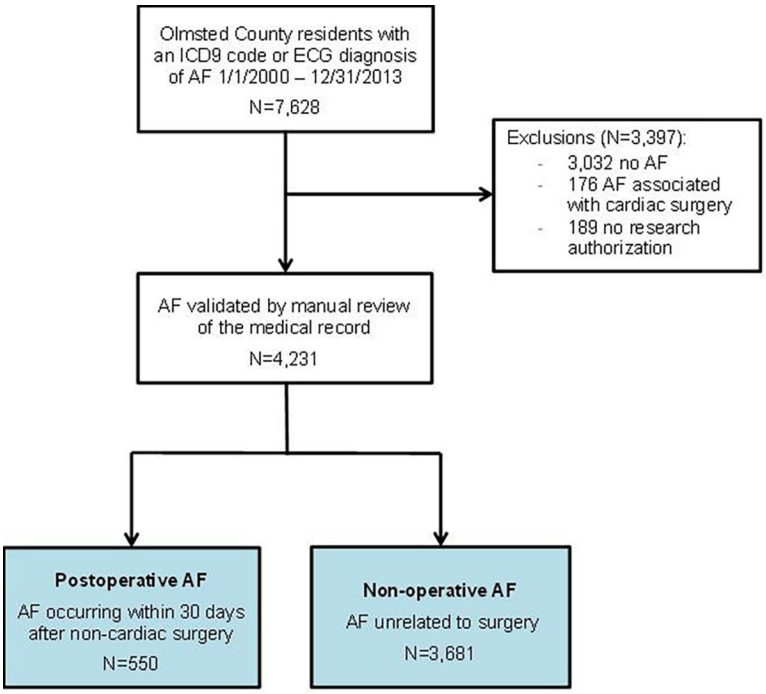
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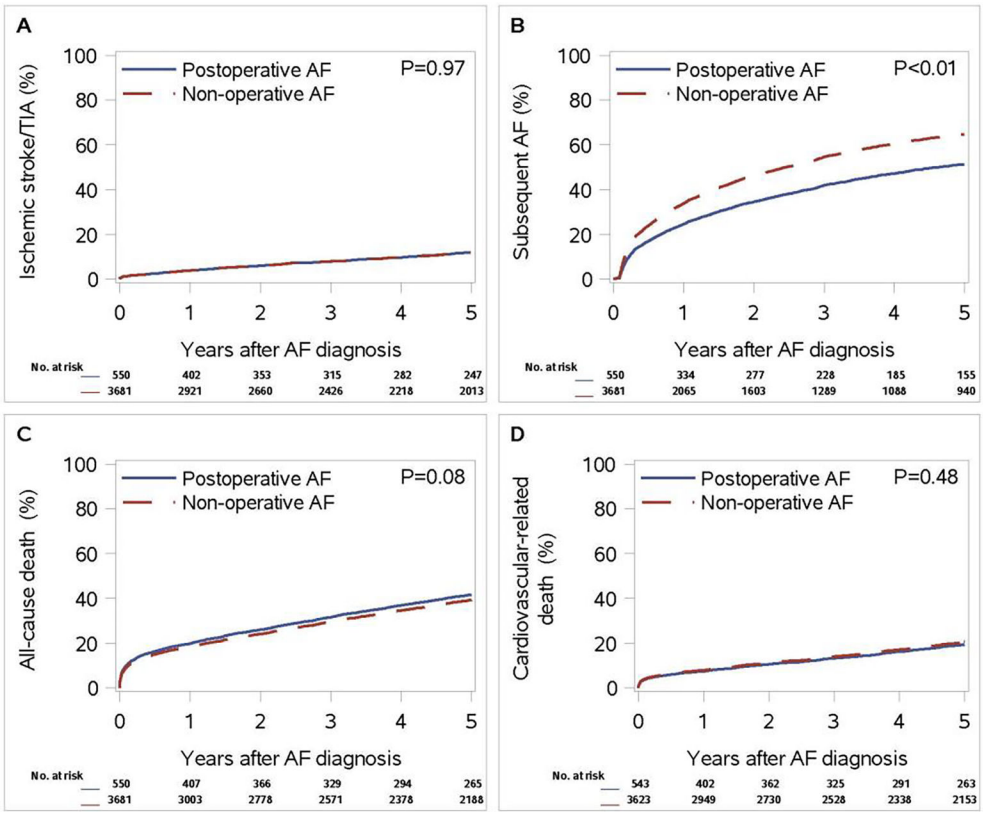
Appendix

Appendix



Appendix Figure 1.
Flow diagram of cohort creation process.

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Appendix Figure 2.

Adjusted cumulative incidence of ischemic stroke or transient ischemic attack (A), subsequent atrial fibrillation (B), all-cause death (C), and cardiovascular-related death (D) comparing patients with postoperative atrial fibrillation and non-operative atrial fibrillation. The curves are weighted using inverse probability weights and adjusted for age, sex, and Charlson comorbidity index.

TIA, transient ischemic attack.

Appendix Table 1.

Results from the propensity score model*

Variable	Coefficient	Standard error
Age	0.0339	0.0280
Age^2	−0.0003	0.0002
Male	−0.5462	0.5549
Age×male	0.0062	0.0075
White race	0.1255	0.2770
Ln(body mass index)	−0.4503	0.2239
Hypertension	0.3301	0.1233
Diabetes	0.2673	0.1159
Heart failure	−0.0717	0.1271

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Variable	Coefficient	Standard error
Myocardial infarction	0.1033	0.1400
Aortic atherosclerotic disease	0.3531	0.1794
Chronic pulmonary disease	0.3332	0.1236
Peripheral vascular disease	0.3474	0.1599
Ischemic stroke/TIA	−0.3220	0.1461
Liver disease	0.6608	0.2760
Renal disease	0.0900	0.1691
Rheumatologic disease	0.4860	0.1851
Malignancy	0.5385	0.1192
Metastatic solid tumor	0.8197	0.1863
Dementia	0.2028	0.2249
Hemiplegia/paraplegia	0.4779	0.2599

*The propensity model excludes 64 patients (62 missing race and 2 missing body mass index).

References

1. Polanczyk CA, Goldman L, Marcantonio ER, Orav EJ, Lee TH. Supraventricular arrhythmia in patients having noncardiac surgery: clinical correlates and effect on length of stay. Ann Intern Med. 1998;129(4):279–85. [PubMed: 9729180]
2. Siontis KC, Gersh BJ, Weston SA, Jiang R, Kashou AH, Roger VL, et al. Association of new-onset atrial fibrillation after noncardiac surgery with subsequent stroke and transient ischemic attack. JAMA. 2020;324(9):871–8. [PubMed: 32870297]
3. Rocca WA, Yawn BP, St Sauver JL, Grossardt BR, Melton LJ 3rd. History of the Rochester Epidemiology Project: half a century of medical records linkage in a US population. Mayo Clinic Proceedings. 2012;87(12):1202–13. [PubMed: 23199802]
4. St. Sauver JL, Grossardt BR, Leibson CL, Yawn BP, Melton LJ 3rd, Rocca WA. Generalizability of epidemiological findings and public health decisions: an illustration from the Rochester Epidemiology Project. Mayo Clinic Proceedings. 2012;87(2):151–60. [PubMed: 22305027]
5. St. Sauver JL, Grossardt BR, Yawn BP, Melton LJ 3rd, Pankratz JJ, Brue SM, et al. Data resource profile: the Rochester Epidemiology Project (REP) medical records-linkage system. Int J Epidemiol. 2012;41(6):1614–24. [PubMed: 23159830]
6. St. Sauver JL, Grossardt BR, Yawn BP, Melton LJ, Rocca WA. Use of a medical records linkage system to enumerate a dynamic population over time: the Rochester Epidemiology Project. Am J Epidemiol. 2011;173(9):1059–68. [PubMed: 21430193]
7. Chamberlain AM, Gersh BJ, Alonso A, Chen LY, Berardi C, Manemann SM, et al. Decade-long trends in atrial fibrillation incidence and survival: a community study. Am J Med. 2015;128(3):260–7. [PubMed: 25446299]
8. Agency for Healthcare Research and Quality. Healthcare Cost and Utilization Project (HCUP): Clinical Classifications Software (CCS) for ICD-9-CM. www.hcup-us.ahrq.gov/toolssoftware/ccs/ccs.jsp accessed February 3, 2022.
9. Deyo RA, Cherkin DC, Ciol MA. Adapting a clinical comorbidity index for use with ICD-9-CM administrative databases. Journal of clinical epidemiology. 1992;45(6):613–9. [PubMed: 1607900]
10. Gage BF, Waterman AD, Shannon W, Boechler M, Rich MW, Radford MJ. Validation of clinical classification schemes for predicting stroke: results from the National Registry of Atrial Fibrillation. JAMA. 2001;285(22):2864–70. [PubMed: 11401607]
11. Lip GY, Nieuwlaat R, Pisters R, Lane DA, Crijns HJ. Refining clinical risk stratification for predicting stroke and thromboembolism in atrial fibrillation using a novel risk factor-based

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approach: the Euro Heart Survey on Atrial Fibrillation. Chest. 2010;137(2):263–72. [PubMed: 19762550]

12. Chamberlain AM, Brown RD Jr., Alonso A, Gersh BJ, Killian JM, Weston SA, et al. No decline in the risk of stroke following incident atrial fibrillation since 2000 in the community: a concerning trend. Journal of the American Heart Association. 2016;5(6):e003408. [PubMed: 27412902]
13. Virani SS, Alonso A, Benjamin EJ, Bittencourt MS, Callaway CW, Carson AP, et al. Heart disease and stroke statistics-2020 update: a report from the American Heart Association. Circulation. 2020;141(9):e139–e596. [PubMed: 31992061]
14. Austin PC. The use of propensity score methods with survival or time-to-event outcomes: reporting measures of effect similar to those used in randomized experiments. Statistics in medicine. 2014;33(7):1242–58. [PubMed: 24122911]
15. D'Agostino RB Jr. Propensity score methods for bias reduction in the comparison of a treatment to a non-randomized control group. Stat Med. 1998;17(19):2265–81. [PubMed: 9802183]
16. Austin PC. An introduction to propensity score methods for reducing the effects of confounding in observational studies. Multivariate Behav Res. 2011;46(3):399–424. [PubMed: 21818162]
17. Hyun J, Cho MS, Nam GB, Kim M, Do U, Kim J, et al. Natural course of new-onset postoperative atrial fibrillation after noncardiac surgery. J Am Heart Assoc. 2021;10(7):e018548. [PubMed: 33739130]
18. Gialdini G, Nearing K, Bhave PD, Bonuccelli U, Iadecola C, Healey JS, et al. Perioperative atrial fibrillation and the long-term risk of ischemic stroke. JAMA. 2014;312(6):616–22. [PubMed: 25117130]
19. Conen D, Alonso-Coello P, Douketis J, Chan MTV, Kurz A, Sigamani A, et al. Risk of stroke and other adverse outcomes in patients with perioperative atrial fibrillation 1 year after non-cardiac surgery. Eur Heart J. 2020;41(5):645–51. [PubMed: 31237939]
20. AlTurki A, Marafi M, Proietti R, Cardinale D, Blackwell R, Dorian P, et al. Major adverse cardiovascular events associated with postoperative atrial fibrillation after noncardiac surgery: a systematic review and meta-analysis. Circ Arrhythm Electrophysiol. 2020;13(1):e007437. [PubMed: 31944855]
21. Butt JH, Olesen JB, Havers-Borgersen E, Gundlund A, Andersson C, Gislason GH, et al. Risk of thromboembolism associated with atrial fibrillation following noncardiac surgery. J Am Coll Cardiol. 2018;72(17):2027–36. [PubMed: 30336826]

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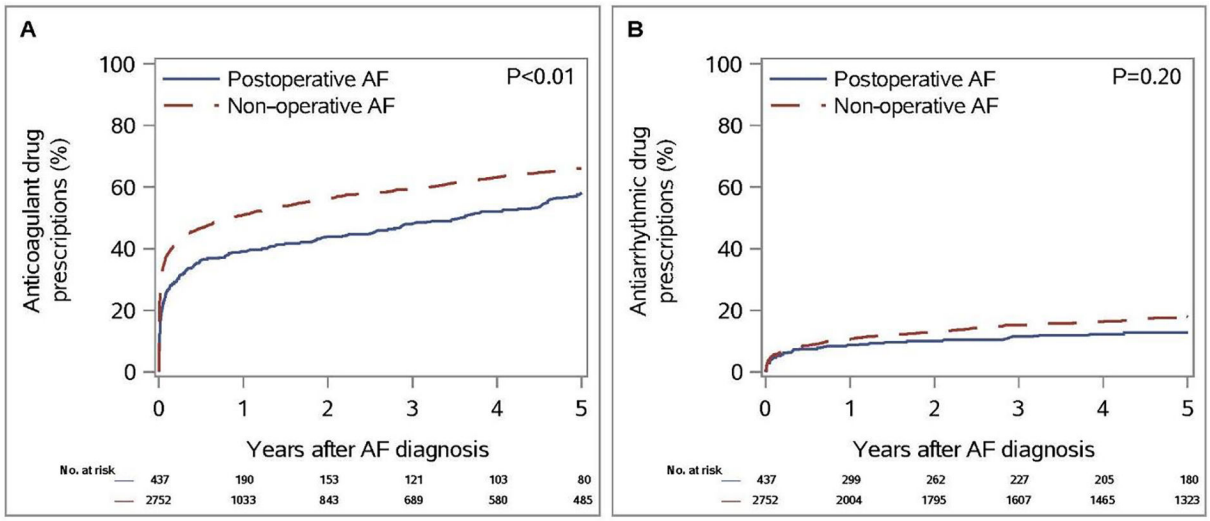


Figure 1.

Cumulative incidence (1 – Kaplan-Meier) of anticoagulant (A) and antiarrhythmic (B) drug prescriptions after the diagnosis of atrial fibrillation comparing patients with postoperative atrial fibrillation and non-operative atrial fibrillation.

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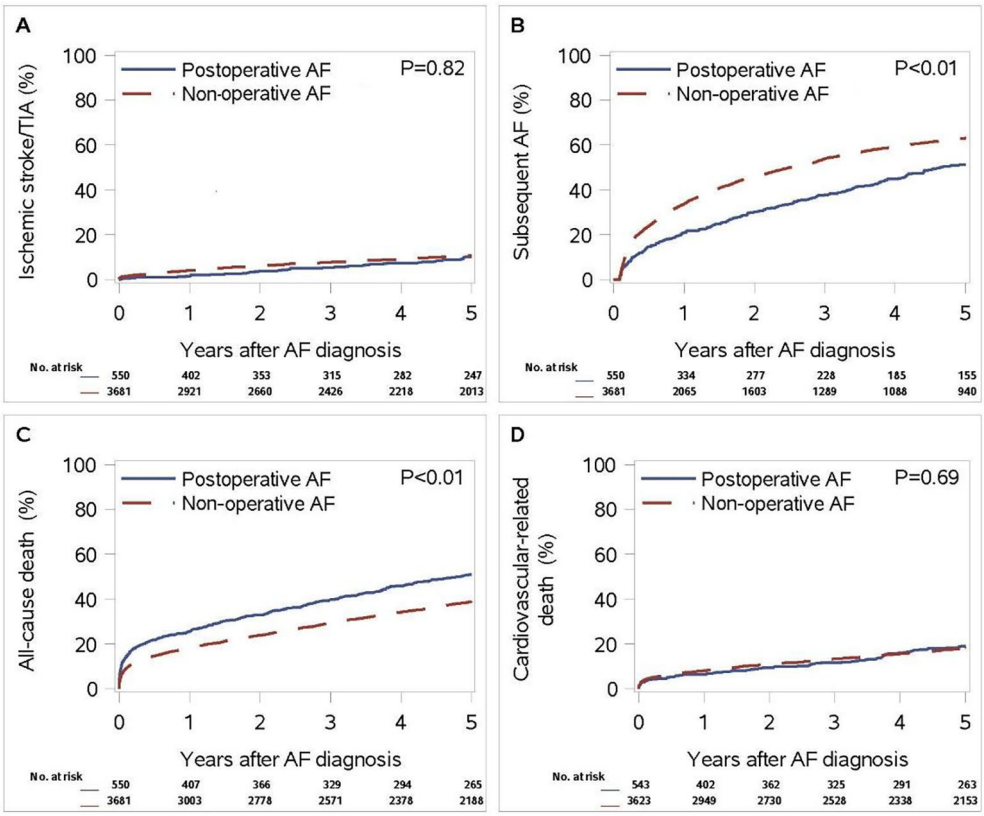


Figure 2.

Cumulative incidence (1 – Kaplan-Meier) of ischemic stroke or transient ischemic attack (A), subsequent atrial fibrillation (B), all-cause death (C), and cardiovascular-related death (D) comparing patients with postoperative atrial fibrillation and non-operative atrial fibrillation.

TIA, transient ischemic attack.

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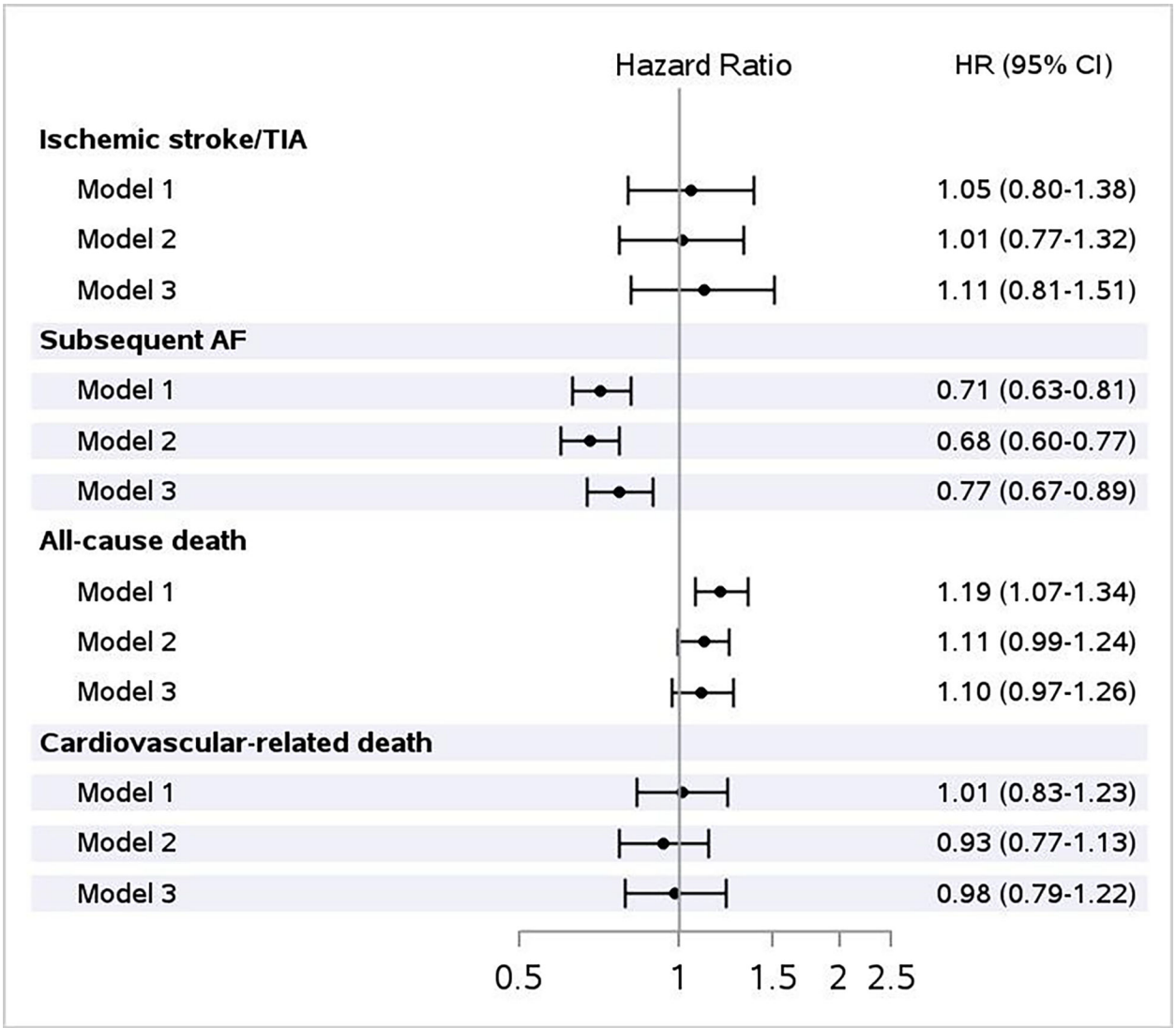


Figure 3.

Hazard ratios (95% confidence interval) for outcomes in patients with postoperative atrial fibrillation and non-operative atrial fibrillation

All models are weighted using inverse probability weights. Model 1 is unadjusted. Model 2 is adjusted for age, sex, and Charlson comorbidity index. Model 3 includes the subset of patients from 2004 forward, and is adjusted for age, sex, Charlson comorbidity index, anticoagulant drug prescriptions, and antiarrhythmic drug prescriptions. The anticoagulant and antiarrhythmic drug prescriptions are modeled as time-dependent variables. 64 people were excluded from Models 1 and 2, and 36 people were excluded from Model 3 due to missing data.

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Table 1.

Baseline characteristics of patients with postoperative atrial fibrillation and non-operative atrial fibrillation

	Original Data			Weighted Data		
	Postoperative AF (N=550)	**Non-operative AF (N=3,681)**	**Standardized difference***	**Postoperative AF (N=533)**	**Non-operative AF (N=3,629)**	**Standardized difference***
Age, years	73.3±13.0	72.3±15.0	0.08	72.6±15.3	72.5±14.7	0.01
Male sex	286 (52.0)	1,933 (52.5)	0.01	275 (51.5)	1895 (52.2)	0.01
Race						
American Indian/Alaskan Native	2 (0.4)	5 (0.1)	0.05	2 (0.3)	5 (0.1)	0.04
Black or African American	3 (0.6)	41 (1.1)	0.06	4 (0.8)	40 (1.1)	0.03
White	527 (97.1)	3508 (96.7)	0.02	516 (96.7)	3514 (96.8)	<0.01
Native Hawaiian/Pacific Islander	0 (0.0)	2 (0.1)	0.03	0.0 (0.0)	2 (0.1)	0.03
Other	5 (0.9)	32 (0.9)	<0.01	6 (1.1)	31 (0.8)	0.03
Asian	6 (1.1)	38 (1.0)	0.01	6 (1.1)	38 (1.1)	<0.01
Missing	7	55		7	55	
Body mass index, kg/m^2						
<25	183 (33.3)	1017 (27.6)	0.12	166 (31.2)	1010 (27.8)	0.07
25 to <30	154 (28.0)	1189 (32.3)	0.09	155 (29.1)	1173 (32.3)	0.07
≥30	213 (38.7)	1473 (40.0)	0.03	212 (39.7)	1446 (39.9)	<0.01
Missing		2			2	
Hypertension	420 (76.4)	2,479 (67.3)	0.20	374 (70.1)	2497 (68.8)	0.03
Diabetes	155 (28.2)	781 (21.2)	0.16	120 (22.4)	802 (22.1)	0.01
Heart failure	125 (22.7)	757 (20.6)	0.05	115 (21.5)	760 (20.9)	0.01
Myocardial infarction	86 (15.6)	458 (12.4)	0.09	70 (13.1)	467 (12.9)	0.01
Aortic atherosclerotic disease	49 (8.9)	186 (5.1)	0.15	35 (6.5)	206 (5.7)	0.03
Chronic pulmonary disease	123 (22.4)	513 (13.9)	0.22	90 (16.9)	551 (15.2)	0.05
Peripheral vascular disease	70 (12.7)	275 (7.5)	0.18	46 (8.6)	297 (8.2)	0.01
Ischemic stroke/TIA	74 (13.5)	519 (14.1)	0.02	79 (14.7)	514 (14.2)	0.02
Liver disease	20 (3.6)	57 (1.5)	0.13	10 (1.8)	67 (1.8)	<0.01
Renal disease	60 (10.9)	277 (7.5)	<0.12	43 (8.1)	292 (8.0)	<0.01

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	Original Data			Weighted Data		
	Postoperative AF (N=550)	**Non-operative AF (N=3,681)**	**Standardized difference***	**Postoperative AF (N=533)**	**Non-operative AF (N=3,629)**	**Standardized difference***
Rheumatologic disease	43 (7.8)	166 (4.5)	0.14	29 (5.4)	184 (5.1)	0.01
Malignancy	183 (33.3)	638 (17.3)	0.37	109 (20.4)	713 (19.7)	0.02
Metastatic solid tumor	67 (12.2)	128 (3.5)	0.33	26 (4.9)	169 (4.7)	0.01
Dementia	27 (4.9)	156 (4.2)	0.03	26 (4.8)	160 (4.4)	0.02
Hemiplegia/paraplegia	22 (4.0)	89 (2.4)	0.09	17 (3.3)	99 (2.7)	0.03
Charlson comorbidity index	3.2±2.8	1.9±2.2	0.50	2.3 ±2.3	2.1 ±2.3	0.08
$CHADS_2$ score	2.1±1.4	1.9±1.4	0.13	2.0 ±1.4	1.9 ±1.4	0.02
CHA_2DS_2-VASc score	3.6±1.8	3.3±2.0	0.18	3.4 ±1.9	3.4 ±2.0	0.05

*Standardized difference <0.10 supports the assumption of balance.

Values presented are n (%) or mean ± standard deviation.

AF, atrial fibrillation; TIA, transient ischemic attack.

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Table 2.

Rates, absolute risks, and absolute risk differences for outcomes in patients with postoperative atrial fibrillation and non-operative atrial fibrillation

			Model 1*		Model 2†		Model 3‡	
	N events	Rate§ (95% CI)	AR (95% CI) at 5 years, %	ARD (95% CI) at 5 years, %‖	AR (95% CI) at 5 years, %	ARD (95% CI) at 5 years, %‖	AR (95% CI) at 5 years, %	ARD (95% CI) at 5 years, %‖
Ischemic stroke/TIA								
Non-operative AF	430	19.5 (17.7–21.4)	10.5 (9.4–11.6)	0.5 (−2.3–3.3)	11.8 (10.5–13.1)	0.1 (−2.9–3.1)	10.7 (9.2–12.1)	1.1 (−2.2–4.4)
Postoperative AF	56	21.0 (15.8–27.2)	11.0 (8.3–13.7)		11.9 (9.0–14.7)		11.7 (8.5–14.8)	
Subsequent AF								
Non-operative AF	2215	188.2 (180.5–196.2)	63.3 (61.5–65.0)	−12.2 (−16.8– −7.6)	64.7 (63.0–66.4)	−13.4 (−17.8– −9.0)	59.4 (57.2–61.4)	−8.8 (−13.7– −3.9)
Postoperative AF	247	133.7 (117.6–151.5)	51.1 (46.6–55.3)		51.3 (46.9–55.4)		50.6 (45.7–55.1)	
All-cause death								
Non-operative AF	2179	91.9 (88.1–95.6)	39.9 (38.3–41.4)	5.6 (1.9–9.3)	39.2 (37.9–40.5)	2.4 (−0.3–5.1)	37.5 (35.9–39.0)	2.3 (−0.8–5.4)
Postoperative AF	386	136.7 (123.4–151.0)	45.5 (41.8–49.0)		41.6 (38.9–44.3)		39.7 (36.7–42.7)	
Cardiovascular-related death								
Non-operative AF	901	38.6 (36.2–41.2)	18.2 (16.9–19.6)	0.2 (−3.0–3.4)	20.4 (19.0–21.7)	−1.1 (−4.0–1.8)	18.2 (16.7–19.7)	−0.3 (−3.4–2.8)
Postoperative AF	116	41.5 (34.3–49.7)	18.4 (15.3–21.5)		19.3 (16.4–22.1)		17.9 (14.8–20.9)	

*Unadjusted. 64 people were excluded due to missing data.

†Inverse probability-weighted model adjusted for age, sex, and Charlson comorbidity index. 64 people were excluded due to missing data.

‡Includes the subset of patients from 2004 forward. Inverse probability-weighted model adjusted for age, sex, Charlson comorbidity index, anticoagulant prescriptions, and antiarrhythmic prescriptions. The anticoagulant and antiarrhythmic prescriptions are modeled as time-dependent variables. 36 people were excluded due to missing data.

§Rate per 1,000 person-years.

‖Absolute risk difference is absolute risk of postoperative atrial fibrillation minus absolute risk of non-operative atrial fibrillation.

AF, atrial fibrillation; AR, absolute risk; ARD, absolute risk difference; CI, confidence interval; HR, hazard ratio; TIA, transient ischemic attack.

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